SAMPLE PAPER 1
Level 1 Functional Skills Mathematics

City & Guilds
A City & Guilds Group Business

Duration: 25 minutes
Total marks: 15

SECTION 1 – CALCULATOR NOT PERMITTED
VERSION 1.1

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

You should have the following for this assessment:
- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler
- a protractor.



You must NOT use a calculator for Section 1.

General instructions
- Read through each question carefully.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.
- You must hand this section in before you can pick up your calculator to begin Section 2.

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Section 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

4 + 3 x 2 =

- BIDMAS rule: Multiplication before addition:
  3 x 2 = 6 --> 4 + 6 = 10

10

[1 mark]

(1 mark)

Q2

25^2 =

- Square number: 25 × 25 = 625

625

[1 mark]

(1 mark)

Q3

What is $\frac{2}{5}$ as a decimal?

- Convert to tenths: 2/5 = 4/10 = 0.4

0.4

[1 mark]

(1 mark)

Q4

–12 + 13 =

- Number line: Start at -12 and move right 13 units: -12 + 13 = 1

(tick one box)

✓ **A** 1 Option A is correct: 1

B – 1

C 25

D – 25

[1 mark]

(1 mark)

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Q5

60% of 300g is

(tick one box)

- 10% of 300g = 30g
- 60% = 6 × 30g = 180g

A 18g ☐

B 50g ☐

C 180g ☑ Option C is correct: 180g

D 200g ☐

[1 mark]

(1 mark)

Q6

155	125	145	95	150	155
125	99	100	178	95	180

- Range = Highest value – Lowest value
  Highest = 180, Lowest = 95 --> 180 – 95 = 85

What is the range of these numbers?

85

[1 mark]

(1 mark)

Q7 A man puts £3000 into a savings account.
The interest rate is 5% per year.

How much will the interest be for the first year?

- 10% of £3,000 = £300 --> 5% = £300 ÷ 2 = £150

£ 150

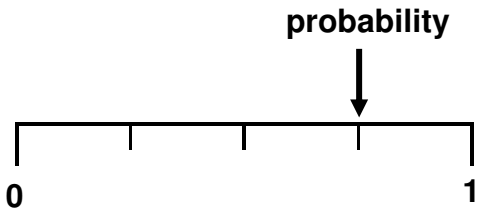
[1 mark]

(1 mark)

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Q8

This scale shows the probability that something will happen.



What probability does the scale show?

(tick one box)

- Pointer is at ~0.85 (between 0.5 and 1, closer to 1)
  This represents a 'likely' event.

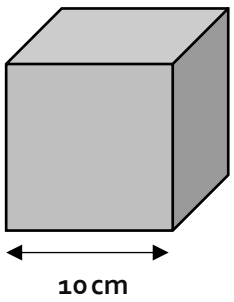
A	impossible	☐
B	certain	☐
C	unlikely	☐
D	likely	☑

Option D is correct: likely

[1 mark]

(1 mark)

Q9



What is the volume of this cube?

- Volume = length × width × height = 10 × 10 × 10 = 1,000 cm³

1000 _____ cm³

[1 mark]

(1 mark)

Q10

$5\frac{2}{5}$ is the same as

- Multiply whole number by denominator and add numerator:
  (5 × 5) + 2 = 25 + 2 = 27 --> 27/5

$\frac{\square}{5}$ → $\frac{27}{5}$

[1 mark]

(1 mark)

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Q11 A customer wants to buy a coat in the sale in a clothes shop. He has £30.



SALE
All coats
Reduced to **70%**
of the ticket price

Step 1: Calculate sale price (70% of £45):
10% of £45 = £4.50
70% = 7 × £4.50 = £31.50 (or 0.70 × £45 = £31.50) [1 mark method]

Step 2: Compare sale price with customer's budget (£30):
Sale price is £31.50, but customer only has £30.00.
£31.50 > £30.00 --> He needs £1.50 more. He cannot afford it.

Can the customer afford to buy the coat?

Yes or No No (Sale price is £31.50)

Show how you got your answer.

[2 marks]
(2 marks)

Q12 A sandwich shop owner makes 1 sandwich with brown bread for every 4 sandwiches he makes with white bread.

Today he needs to make 600 sandwiches altogether.

- Total parts in ratio = 1 + 4 = 5 parts

How many sandwiches should he make with brown bread today?

- Value of 1 part = 600 ÷ 5 = 120 sandwiches
- Brown sandwiches = 1 part = 1 × 120 = 120

120 **sandwiches**

[1 mark]
(1 mark)

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Q13 Jon is a checkout assistant in a supermarket.

There are 5 tills and 5 assistants. The supervisor allocates the tills to the assistants randomly at the start of each day.

No one likes the till next to the door.

What is the probability that Jon will get the till next to the door today?
Give your answer as a fraction in its simplest form.

- Number of favourable outcomes = 1 (till next to door)
- Total possible outcomes = 5 tills
- Probability = 1/5 (or 0.2 / 20%)

Answer: 1/5

$\frac{\square}{\square}$

[1 mark]

(1 mark)

Q14 A van has a fuel tank that holds 60 litres when full.

This diagram shows the fuel gauge on the van.



Approximately how many litres of fuel are left in the tank? **Give your answer in whole litres.**

- Gauge is at the 1/4 mark (halfway between Empty and Half):
  60 litres ÷ 4 = 15 litres remaining

15 **litres**

[1 mark]

(1 mark)

End of Section 1.
When you have finished you MUST hand this booklet in to the invigilator before you can pick up your calculator to start Section 2.

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SAMPLE PAPER 1
Level 1 Functional Skills Mathematics

Duration: 1 hour 20 minutes
Total marks: 45

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

SECTION 2 – CALCULATOR PERMITTED
VERSION 1.1

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

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- an eraser
- a 30cm ruler
- a protractor.



You may use a calculator for Section 2.

General instructions

- Read through each question carefully.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages **including graph paper** at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks available in this section.

You should check all your work as you go along.

You may use a calculator.



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Q1

Write **eight hundred and seven thousand, two hundred and five** in figures.

• 800,000 + 7,000 + 200 + 5 = 807,205

807 205

[1 mark]

(1 mark)

Q2

What is 2 043.666666 rounded to 2 decimal places?

• Column addition: 2034.70 + 8.97 = 2043.67

2043.67

[1 mark]

(1 mark)

Q3

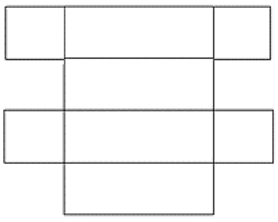
Which one of these nets will fold to make a cube?

• Shape D (parallelogram) has rotational symmetry of order 2 (180° rotation)

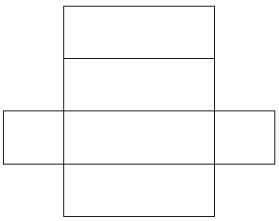
Answer: D

[1 mark]

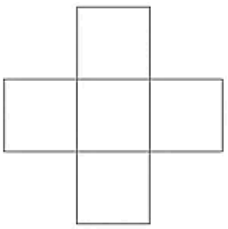
A



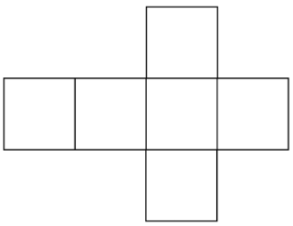
B



C



D



(tick one box)

☐ ☐ ☐ ☐

(1 mark)

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Q4

Which one of these fractions is the **biggest** number?

(tick one box)

- ☐ A $\frac{4}{10}$
- ☑ B $\frac{3}{4}$
- ☐ C $\frac{2}{5}$
- ☐ D $\frac{1}{3}$

• Metres (m) is standard unit for athletic pitches

Option B is correct: m

[1 mark]

(1 mark)

Q5

Which one of the following lists is in decreasing order from the largest to the smallest?

(*tick one box*)

- ☐ **A** 252,080 252,300 250,900
- ☐ **B** 252,080 250,900 252,300
- ☑ **C** 252,300 252,080 250,900
- ☐ **D** 252,300 250,900 252,080

• Compare place values: 0.990 > 0.909 > 0.900 > 0.090

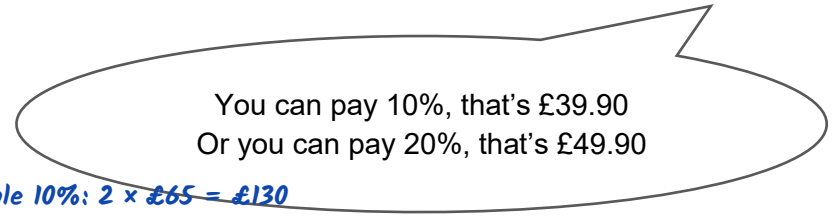
Option C is correct: 0.99

[1 mark]

(1 mark)

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Q6 A customer wants to buy a sofa. The salesperson says he must pay a deposit.



- 10% of £650 = £65
- 20% should be double 10%: 2 × £65 = £130

Explanation: The salesperson only calculated 10% instead of 20%.
The 20% deposit should be £130, which is double £65.

Explain why the salesperson must have made a mistake.

[1 mark]

Explanation

(1 mark)

Step 1: Calculate how many 500g batches can be made with 13 eggs:

Q7 13 ÷ 4 = 3.25 batches

A gardener needs to order fertiliser for a piece of land.
The piece of land is a square with sides measuring 8 metres.

3.25 × 500g = 1,625g [M1]

Step 2: Calculate total weight of cake mix needed in grams and kg:

This formula shows how many **grams** of fertiliser she needs.

Convert to kg: 1,625 ÷ 1,000 = 1.625 kg (or 1.6 kg) [A1]

grams of fertiliser needed = length in metres x width in metres x 25

Step 3: Select appropriate bag size (available: 1kg, 2kg, 5kg):

The supplier sells these bags of fertiliser.

1kg is not enough (1.625kg needed). 5kg is unnecessarily large.
Therefore, the 2kg bag provides enough mix with minimal waste.

Answer: 2 kg bag (needs 1.625 kg)



1 kilogram **2 kilograms** **5 kilograms** **10 kilograms**

Which bag of fertiliser should the gardener order? Include figures to explain your answer.

Show all your working.

[4 marks]

_____ **kg bag**

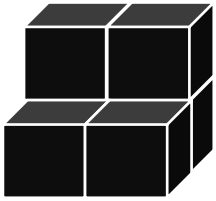
Explanation

(3 marks)

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Q8

A shelf stacker is making a display in a shop. It is made from six identical boxes. Each box is a cube.



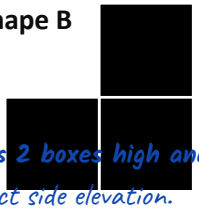
side X

Which one of the shapes below is a side view (elevation) of the display from **side X** ?

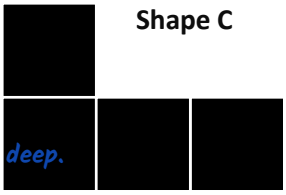
Shape A



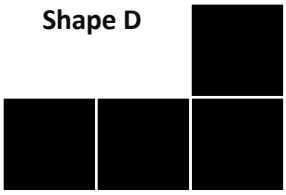
Shape B



Shape C



Shape D



Part a: Side Elevation View:

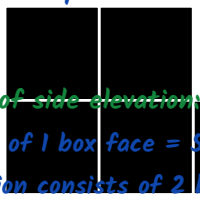
From the side, the display is 2 boxes high and 1 box deep.

Option B represents the correct side elevation.

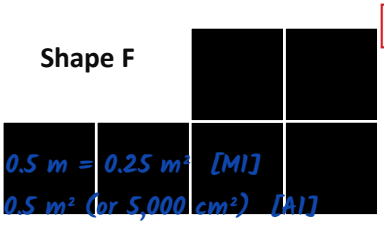
Side elevation option: B

[1 mark]

Shape E



Shape F



Part b: Area of side elevation:

Dimensions of 1 box face = 50 cm × 50 cm = 0.5 m × 0.5 m = 0.25 m² [M1]

Side elevation consists of 2 box faces: 2 × 0.25 m² = 0.5 m² (or 5,000 cm²) [A1]

Area: 0.5 m² (or 5,000 cm²)

[2 marks]

The side view (elevation) from side X is **Shape** _____

Part c: Total weight of display:

Each box has sides measuring **0.5m**

Display is made from 6 identical boxes, each weighing 950g.

Work out the area of this side of the display. **Give units in your answer.**

Total weight = 6 × 950g = 5,700g [M1]

Show all your working.

[2 marks]

Convert to kg: 5,700 ÷ 1,000 = 5.7 kg [A1]

Total weight: 5.7 kg

Area _____

Each box weighs 950g

What is the weight of the whole display in **kilograms**?

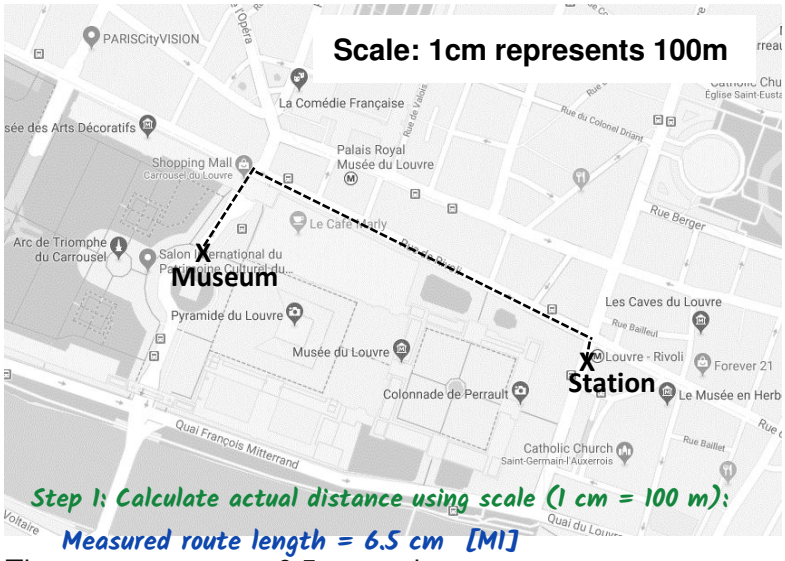
Show all your working.

_____**kg**

(4 marks)

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Q9 A tourist wants to know how long it will take to walk to a museum.
This map shows the route from a nearby station to the museum.



Step 1: Calculate actual distance using scale (1 cm = 100 m):

Measured route length = 6.5 cm [M1]

The route measures 6.5cm on the map.

Actual distance = 6.5 × 100 m = 650 m = 0.65 km [A1]

It takes the tourist 10 minutes to walk a kilometre.

Step 2: Calculate walking time (average walking speed = 6 km/h = 100 m/min):

Time = 650 m ÷ 100 m/min = 6.5 minutes [M1]

How long will it take him to walk to the museum from the station?

Give units in your answer.

0.5 minute = 30 seconds --> 6 minutes 30 seconds [A1]

Answer: 6.5 minutes (or 6 minutes 30 seconds)

[4 marks]

Show all your working.

Amount of time to walk _____

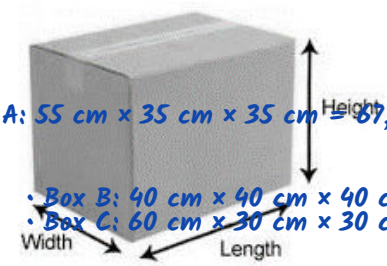
(4 marks)

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Q10 A sales rep needs some cardboard boxes to store samples at work.

The office supplier has three different sized boxes.

Step 1: Calculate volume for each box (Volume = l × w × h):



Box	Width	Length	Height
A	35cm	35cm	55cm
B	40cm	40cm	40cm
C	30cm	60cm	30cm

- Box A: 55 cm × 35 cm × 35 cm = 67,375 cm³ [M1]
- Box B: 40 cm × 40 cm × 40 cm = 64,000 cm³ [A1]
- Box C: 60 cm × 30 cm × 30 cm = 54,000 cm³ [A1]

Step 2: Compare volumes to identify largest capacity:

67,375 cm³ (A) > 64,000 cm³ (B) > 54,000 cm³ (C)

Box A holds 3,375 cm^3 more than Box B and 13,375 cm^3 more than Box C.

Which box has the largest volume?

Decision: Box A has the largest volume (67,375 cm³)

Explain your answer. Include figures to support your explanation. [4 marks]

Answer _____

Show all your working.

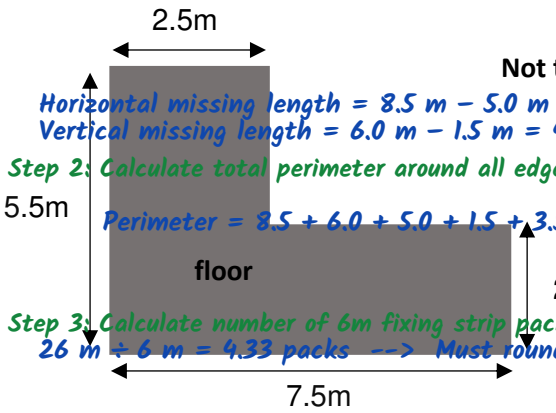
Explanation

(4 marks)

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Q11 A carpet fitter needs to order some fixing strips to hold the carpet all around the edges of this floor.

Step 1: Find missing lengths of L-shaped room:



Horizontal missing length = 8.5 m − 5.0 m = 3.5 m [M1]
Vertical missing length = 6.0 m − 1.5 m = 4.5 m

Step 2: Calculate total perimeter around all edges:

Perimeter = 8.5 + 6.0 + 5.0 + 1.5 + 3.5 + 4.5 = 26 m [A1]

Step 3: Calculate number of 6m fixing strip packs needed:

26 m ÷ 6 m = 4.33 packs --> Must round up to 5 whole packs [M1]

Step 4: Calculate total cost with 'Buy 3 get 2 free' offer (£7.99 per pack):

He can cut and join fixing strips.

Needs 5 packs: Buys 3 packs and gets 2 packs free! [M1]

Cost = 3 × £7.99 = £23.97 [A1]

Answer: 5 packs needed. Total cost £23.97

[5 marks]

Carpet Supplies Direct

Fixing strips: 6-metre packs
£7.99 per pack

*** Buy one get one free ***

Work out the total cost for the packs of fixing strips the carpet fitter needs to order.

Show all your working.

Total cost £_____

(5 marks)

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Q12 A doctors' receptionist wants to display a notice to show patients how long, on average, they will wait to see a doctor.

The receptionist recorded how long past their appointment time patients at the surgery waited to see their doctor yesterday morning.

Step 1: Calculate sum of all waiting times:

Sum = 5 + 4 + 8 + 12 + 4 + 7 + 6 + 5 + 3 + 6 = 60 minutes [M1]

Minutes waiting (to the nearest minute)		
4	8	6
4	5	3
10	7	5
9	7	4

Step 2: Calculate mean waiting time (Total ÷ 10 patients):

Mean = 60 ÷ 10 = 6 minutes [A1]

Step 3: Tutor comment on displaying this notice to patients:

Complete the notice.

Yesterday's sample was only 10 patients and may not be typical.
The clinic should track times over several days for a reliable average. [A1]

Show all your working.

Average wait: 6 minutes

[3 marks]

Hilltop Surgery

Average waiting time

You should expect to wait ☐ **minutes to see a doctor today.**

Explain why using the waiting times for yesterday morning might not give a suitable waiting time to put on the notice.

Explanation

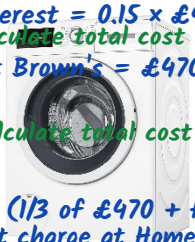
(3 marks)

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Q13 A customer needs to buy a washing machine. He wants to pay monthly.

Two different shops have the washing machine he wants.

BROWNS



Price £470

Pay monthly offer:

Interest is 15% of the price.

Pay the total amount in 10 equal monthly instalments. First instalment must be paid today.

HS

Home Superstore



Price £525

Pay monthly offer:

Interest is 5% of the price.

Pay 1/3 today and the rest in 4 equal monthly payments.

15% interest = 0.15 × £470 = £70.50 [M1]

Step 1: Calculate total cost at Brown's (Cash price £470 + 15% interest):

Total at Brown's = £470 + £70.50 = £540.50 (or 10 × £54.05) [A1]

Step 2: Calculate total cost at Home Supply (HS):

Deposit (1/3 of £470 + fee): Total with credit = £551.25 [A1]

(Interest charge at Home Supply is £26.25)

Recommendation: Buy from Brown's to save £10.75. [A1]

Step 3: Compare both options and calculate savings:

Difference = £551.25 − £540.50 = £10.75 cheaper at Brown's [M1]

Brown's total: £540.50 · Home Supply total: £551.25

Decision: Brown's is cheaper by £10.75

[5 marks]

Work out which offer is cheaper and by how much. How much would the customer need to pay today?

Show all your working.

Shop_____ offer is cheaper by £_____

Amount to pay today £_____

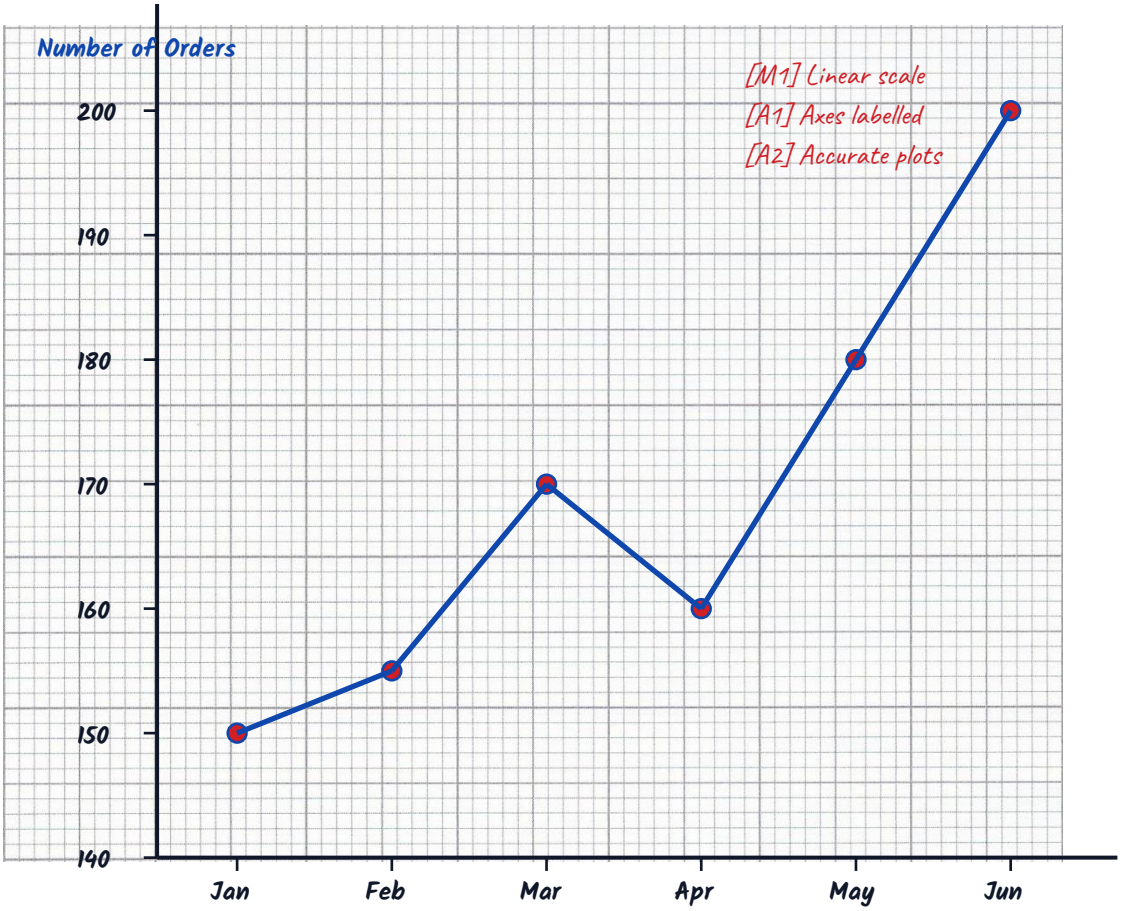
(5 marks)

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Q14 This table shows the number of orders a sales person got in six months.

Month	Number of orders
January	150
February	155
March	170
April	160
May	180
June	200

Draw a line graph to show this information.



Explain what your graph shows about the number of orders.

Explanation

Explanation of Trend:

The graph shows that orders are generally increasing over the six-month period, rising from 150 in January to 200 in June, with a temporary dip in April. [A1]

[5 marks]

(5 marks)

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Q15 Your boss needs to go to a meeting in London on Sunday.
He lives half an hour drive from Stockport station.
He doesn't want to leave home too early.

The meeting starts at 14:00. It is in a hotel a 15-minute walk from the station in London.

Step 1: Work backwards from meeting time (arrive hotel by 14:00):

This is the train timetable for Sunday.

15 minutes from Euston to hotel --> Must arrive London by 13:45 at latest. [M1]

Sundays						
Manchester	0920	1020	1035	1113	1135	1155
Stockport	0928	1029	1046	1124	1144	1205
Stoke-on-Trent	1000	1100	1115	1152	1214	1235
London	1206	1254	1257	1328	1347	1410

Step 2: Select appropriate train from timetable:
Arrival at hotel = 13:28 + 15 mins = 13:43 (comfortably on time!).
Train departing Stockport at 11:24 arrives London Euston at 13:28 [M1]

Step 3: Departure time from home (30 mins drive to station):
11:24 - 30 minutes = 10:54 am departure from home. [A1]
(Accept 10:46 train departing Stockport arriving London 12:57, hotel 13:12)

The website also has information about ticket prices.

Step 4: Calculate total journey cost:
Train ticket (£175.00) + Tube (£4.55) + Station parking (£40.00) [M1]

Prices between Stockport and London		
Ticket type	**Off-peak (weekends)**	**Peak (any day)**
Single (one way)	£44.55	£175.00
Return (two journeys)	£89.10	£242.00

Total = £175.00 + £4.55 + £40.00 = £219.55 [A1]

Schedule: Leave home at 10:54 · Catch 11:24 train · Arrive hotel at 13:43

[4 marks]

Total Journey Cost: £219.55

Work out the best train for your boss to catch.
Explain why you chose that train.

Show your working.

Train _____

Explanation

Your boss will travel back on Monday

Work out the cheapest cost for your boss to travel to London and back.

Show your working.

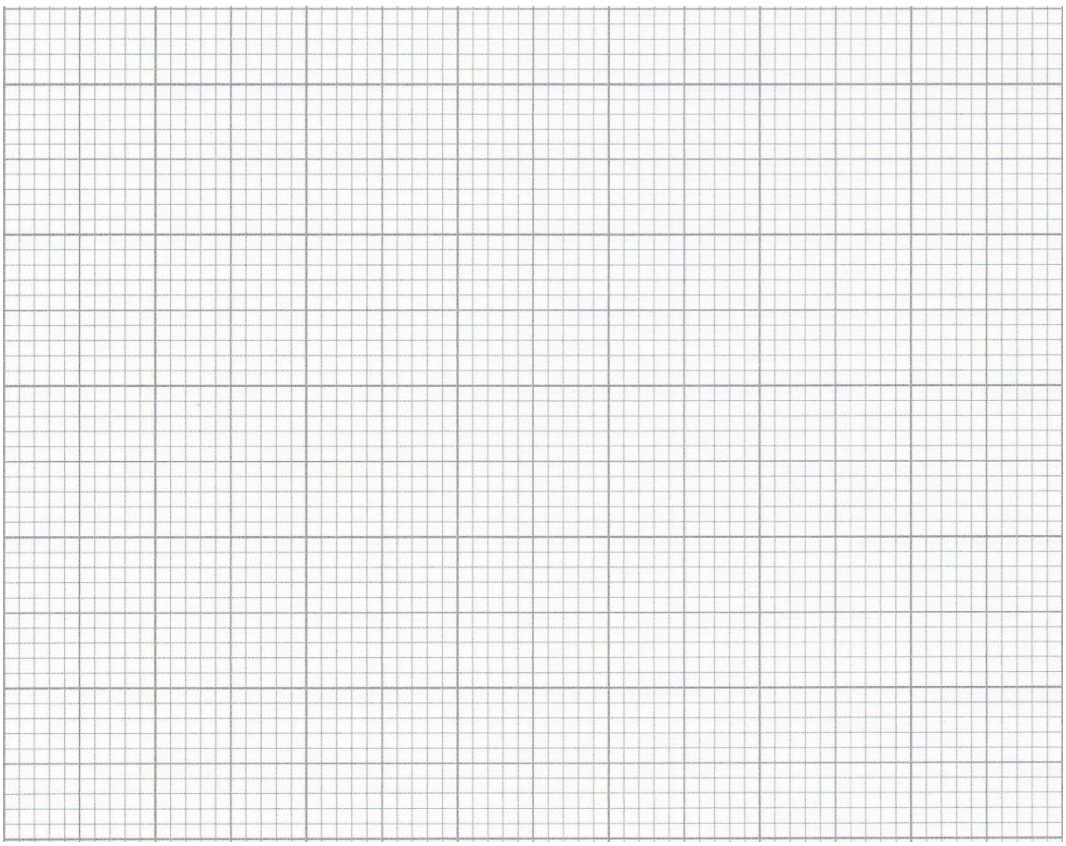
Cheapest cost £ _____

(6 marks)

End of Section 2

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Spare graph paper for Question 14



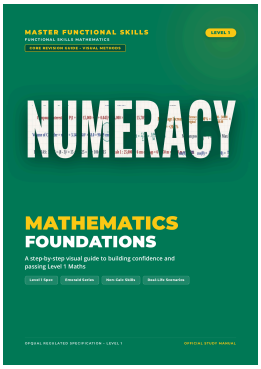
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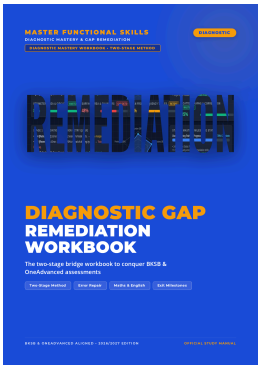


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